Chirality from $(\mathbb{S}, \tau)$: Gauge Structure, Forced Asymmetry, and the Protected Mirror

Harald Rønning

Revision 1.0 — closed. This paper consolidates Papers I–V of this corpus (the unified paper, the Phase 3, 4, and 5 reports, and the seam note) into a single, standalone technical account of the chirality program: from the algebraic seed to the exact gauge structure it forces, to the domain-wall dynamics that makes that structure chiral, to the theorem proving nothing derived can select which mirror the chirality faces, to the combinatorial and dynamical structures that can *carry* that choice without ever supplying it. Every claim below is inherited from Papers I–V, each independently verified in two implementations there; no new computation is performed here. Papers I–V remain the detailed working record, including the two rounds of hostile review, the error ledgers, and the development notes; this paper is the citable reference for the finished result.

Abstract

A single algebraic object — the mediator direction fixing a complex structure on the imaginary octonions, and the idempotent it determines in the exceptional Jordan algebra $J_3(\mathbb{O})$ — forces the Standard Model's gauge group with exact hypercharges and zero free parameters. The same algebra's own dynamics, applied to a domain-wall construction on this matter content, forces a chiral wall sector under the unbroken gauge group: a lepton singlet and a quark triplet, necessarily not self-conjugate, by Schur's lemma alone. This leaves exactly four gauge-compatible orientation structures — a Standard-Model-shaped class and its mirror — with the dynamics provably indifferent between them. A four-part protection theorem then closes every channel the framework's own derived operators could use to select one mirror over the other, including the framework's one genuinely derived sign, the thermal arrow. The sole remaining candidate is the algebra's own settling dynamics, and the only structure of $\mathbb{S}$ capable of hosting it is the zero-divisor locus — the one place the algebra's two Cayley–Dickson halves are forced to mix. Mapping this locus combinatorially into seven "box-kites" and testing the resulting dynamics to a conclusion: the mechanism *can* carry the mirror bit, faithfully, through an exact and fully verified chain — but nothing in the framework supplies which point of the locus anchors it. The chirality is forced; its orientation is protected from every derived mechanism; the one thing capable of carrying that orientation is real, mapped, and tested — and the framework hands it no address.

---

1. Introduction and overview

This paper follows one question through five stages, each closed and independently verified before the next began.

Stage 1 (§3): is there a gauge structure at all, and are its charges forced? Starting from the mediator direction in the octonions and the single idempotent it determines in $J_3(\mathbb{O})$, the Standard Model's gauge group $SU(3) \times SU(2) \times U(1)$ emerges with every hypercharge ratio fixed by representation theory alone — no phenomenological input, and one prediction (a forbidden u_R-type slot at a specific charge) confirmed before the computation that tested it was run.

Stage 2 (§4): does this gauge structure force a chiral sector? Building a domain-wall construction on the matter module using the framework's own derived dynamics, the localized wall content is forced to be chiral — not by tuning, but by Schur's lemma acting on an irreducible color triplet that cannot be self-conjugate. This also reveals that the construction admits exactly four gauge-compatible orientation structures, one Standard-Model-shaped and three mirror or mixed variants — and the dynamics that forces the chirality is blind to all four.

Stage 3 (§5): can anything the framework derives choose among the four? No. A closed, four-part partition of the entire legal operator algebra shows every derived operator either vanishes on the wall, has provably zero alignment with the orientation, is a structureless per-line scalar, or carries a sign that is itself a gauge or convention choice rather than a derived fact. The framework's one genuinely derived sign — the thermal arrow — is shown to be trapped in the same protected sector at every perturbative order. The orientation is a theorem-grade initial condition, not a gap.

Stage 4 (§6): where, if anywhere, could a non-derived mechanism live? The only structure of $\mathbb{S}$ that necessarily mixes both Cayley–Dickson halves — and hence the only place a genuinely new, non-quasi-free dynamics could be native to the algebra — is the zero-divisor locus. Mapped combinatorially, this locus organizes into seven “box-kites” of twelve elements each, and an exact automorphism (read, on independently verified facts, as the algebra's own PT operation) splits these seven into three orientation-blind kites and four orientation-bearing ones.

Stage 5 (§7): does this structure actually carry the orientation, and does anything select it? Testing a specific, textually-sourced candidate mechanism to its conclusion: the construction bifurcates into direction-free members (contentless for orientation, though not structureless) and anchored members, which carry the mirror bit faithfully through a fully verified chain to the gauge side. But no computation, theorem, or textual source anywhere in the corpus supplies *which* point of the zero-divisor locus anchors the mechanism. Settling can carry the orientation. It cannot author it.

§8 states the complete arc as one narrative; §9 catalogues, precisely, everything this leaves open; §10 is a note on the verification discipline that produced every claim above.

---

2. The algebraic seed

The tower and the split. $\mathbb{S}$ is the sedenions, the fourth Cayley–Dickson doubling of $\mathbb{R}$ ($\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S}$), sixteen-dimensional, decomposing as $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$: the octonions and a second copy twisted by the doubling. τ is the involution negating the odd (second-copy) half. The octonion units are $e_0, \dots, e_7$, with e_0 the real unit; e_4 is singled out as the **mediator** — the unique imaginary direction every derivation of $\mathbb{O}$ fixes ($D(e_4) = 0$ for all $D \in \mathfrak{g}_2 = \mathrm{Der}(\mathbb{O})$) — splitting the remaining six imaginary units into a triple $P = \{e_1, e_2, e_3\}$ and a triple $X = \{e_5, e_6, e_7\}$.

Two standing facts, both elementary. Neither $\mathbb{S}$ nor the exceptional Jordan algebra $J_3(\mathbb{O})$ (27-dimensional Hermitian 3×3 octonionic matrices, the other object this program depends on) embeds in the other — dimension prevents one direction, commutativity the other. $J_3(\mathbb{O})$ is adopted from the classical literature (Jordan–von Neumann–Wigner, 1934), not derived from $\mathbb{O}$; the bridge between the two constructions operates at the level of symmetries, $\mathfrak{g}_2 \subset \mathfrak{f}_4 = \mathrm{Der}(J_3(\mathbb{O}))$, and — as the bridge theorem of §3 shows — at the level of one shared module.

The mediator's idempotent. Writing $A := (0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}e_4, 0, 0)$ (in the standard 3×3 Hermitian coordinates), A is a primitive idempotent of $J_3(\mathbb{O})$: Peirce spectrum $\{0, \frac{1}{2}, 1\}$ at multiplicities $\{10, 16, 1\}$. Both objects — the mediator in $\mathbb{O}$, the idempotent it determines in $J_3(\mathbb{O})$ — are the entire input to the gauge derivation that follows.

---

3. Gauge structure and exact charges

3.1 Color and electroweak, as two commutants

$\mathfrak{g}_2$ is 14-dimensional; the mediator's stabilizer within it is exactly 8-dimensional, acting as the genuine fundamental of $\mathfrak{su}(3)$ on the six non-mediator imaginary directions (irreducible by Schur, commutant dimension 1). Embedded in $\mathfrak{f}_4$ (52-dimensional, both directly constructed and verified as a derivation algebra of $J_3(\mathbb{O})$), this **is** the published exceptional-Jordan program's color $\mathfrak{su}(3)$ — the stabilizer, within G_2, of a fixed unit imaginary octonion, arrived at here independently.

The centralizer of color within $\mathfrak{f}_4$ is exactly 8-dimensional, closed, negative-definite: a second, independent $\mathfrak{su}(3)$, the **electroweak** $\mathfrak{su}(3)$. Adjoining the idempotent frame's traceless Jordan multiplications to $\mathfrak{f}_4$ closes to $\mathfrak{e}_{6(-26)}$ (Killing signature exactly -26, verified from independently assembled structure constants). A's stabilizer within the electroweak $\mathfrak{su}(3)$ is exactly 4-dimensional: a 1-dimensional center $\mathfrak{u}(1)_Y$ (commuting with color to 10^{-16} — hypercharge is color-blind) and a 3-dimensional derived $\mathfrak{su}(2)_{\text{weak}}$.

3.2 Three branching theorems and the exact census

Theorem (ratio 3). Under $\mathfrak{su}(3) \supset \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with $Y = \text{diag}(a, a, -2a)$, the adjoint's doublets carry charge $\pm 3a$ against the fundamental doublet's a — measured first, proven after: the Standard Model's own lepton-to-quark doublet hypercharge ratio, -3, is this Lie-theoretic number.

Theorem (8/1/7, pre-registered). $26 = (\mathbf{1}, \mathbf{8}) \oplus (\mathbf{3}, \mathbf{3}) \oplus (\bar{\mathbf{3}}, \bar{\mathbf{3}})$ forces exactly 8 doublets, 1 triplet, 7 singlets, and predicts — in exact, stated form before the computation ran — a color singlet at $Y = 0$, colored singlets at ratio -2, and a state at ratio $+4$ forbidden. All confirmed, including the required absence.

Theorem (Peirce grade is weak type). By F_4-transitivity on primitive idempotents, a state is a weak doublet iff its Peirce grade carries exactly one index matching the weak $\mathfrak{su}(2)$'s own rotated pair — verbatim the Peirce-$\frac{1}{2}$ condition, verified three independent ways.

The census. 6 colored-doublet pairs, 2 lepton-doublet pairs (ratio ± 3), 3 colored-singlet pairs (ratio ∓ 2), one sterile singlet, one neutral triplet: 26 states, every multiplicity and ratio a branching consequence, not a fit. With $T_3 = \pm\frac{1}{2}$ on doublets and one calibration ($Y_{\text{colored doublet}} = \frac{1}{6}$), $Q := T_3 + Y$ lands every state on a Standard Model value, $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$.

3.3 Chirality's first appearance: $\mathbb{O}e_8$'s Clifford structure

$\mathbb{O}e_8$ carries, under $\mathbb{S}$'s own multiplication, a uniform negative-definite Clifford structure across all eight directions: $\mathbb{S}$ supplies Cl$(0, 8)$. Eight generators (not seven — odd Clifford algebras cannot supply a chirality operator at all) give a genuine chirality operator Ω, $\Omega^2 = +1$, every generator swapping its eigenspaces exactly. The mechanism is evenness, not signature. Each half carries the 8-dimensional spinor of $\mathfrak{spin}(7)$; the full even symmetry $\mathfrak{so}(8)$ exhibits the complete, textbook triality between the two half-spinors and the vector representation — pairwise inequivalent, sharing every quadratic invariant.

3.4 The bridge theorem and $T_3 = \lambda J$

Stab(A) is exactly 36-dimensional ($=$ dim Spin(9)) and acts irreducibly on the Peirce-$\frac{1}{2}$ module $J_{\frac{1}{2}}$. The nine traceless Peirce-0 directions satisfy a native Clifford relation on $J_{\frac{1}{2}}$; under an explicit dictionary $\varphi : \mathbb{O} \to$ (gauge-side slot), the composite operator J is a genuine complex structure commuting with color, and the color-centralizer of the gauge side's even algebra matches the $\mathbb{O}e_8$ side's forced centralizer exactly. **Same module, same even structure, opposite real signature.**

This yields the identity $T_3 = \lambda J$ exactly ($\lambda^2 = \frac{1}{24}$): weak isospin's own Cartan generator is a multiple of the bridge's transported chirality operator — one object, reached independently from the electroweak stabilizer chain and from the $\mathbb{O}e_8$ Clifford dictionary. $J_{\frac{1}{2}}$ is further shown to be a quaternionic module $\mathbb{H}^4$, with $\mathfrak{su}(2)_{\text{weak}}$ acting as $\mathfrak{sp}(1)$ and T_3's transported chirality operator as the quaternionic i.

The mechanical fact, held open by design: weak isospin acts **across** the transported chirality split; hypercharge acts **along** it. Whether the weak sector merely contains a chirality axis (Reading A) or whether a further, Lorentz-anchored γ^5-analogue is a separate object (Reading B) is not decided in Paper I; the two readings are recorded at equal rank, with Reading B's evidence heavier by the time Papers II–V complete the picture (§4 onward all adopt Reading B explicitly, as the internal-module reading pending a spacetime program).

3.5 The rank-6 torus and the census's boundary

Every one of the six generators of $\mathfrak{e}_6$'s Cartan torus is a named function of the mediator and the idempotent frame — no unnamed generator remains. On the complexified 27-dimensional census (adding two further charges Q_A, Q_w), twelve conjugate pairs are related by a proven symmetric/antisymmetric split, plus three self-conjugate states that are the idempotent frame itself. Exact-arithmetic analysis of a modified hypercharge $Y' = \alpha Y + \beta Q_A + \gamma Q_w$ shows that reaching a u_R-type slot at $\alpha = \frac{3}{2}$ costs exactly two ratio units, paid by whichever sector is not held fixed; the e_R-type slot (color-singlet, weak-singlet) is **structurally absent** from the census — not a gap in the construction, but an exact feature of what $J_3(\mathbb{O})$'s own 27-dimensional content contains.

A separate theorem shows the resulting embedding $G_{SM} \hookrightarrow E_6$ is **inequivalent** to the published $SO(10)$-GUT embedding — three computed witnesses, including distinct branching multisets for two conjugate copies of $\mathfrak{su}(2)$ inside one and the same $\mathfrak{so}(10)$. This locates, precisely, where the published exceptional-Jordan program's further extensions become necessary rather than optional; it does not compete with that program, which this construction's gauge sector reproduces independently.

4. Forced chirality

4.1 The four orientations

On the 16-dimensional matter module, the real commutants of the gauge chain have dimensions 24/16/4 (color / color+Y / full G_{SM}). Exactly **four** G_{SM}-compatible complex structures exist — $\pm J_L \oplus \pm J_Q$ — and every one selects a chiral $Q \oplus L$ sector, splitting $2+2$ into a Standard-Model-shaped class (lepton/quark hypercharge ratio -3) and a mixed class ($+3$). This is a corollary of a general classification theorem: on an isotypic gauge block, compatible complex structures are classified by signature, with chirality equivalent to definite signature equivalent to centrality in the block commutant.

4.2 The domain wall and the gate theorem

Constructing a fermionic dynamics on the module via a graded contraction of $\mathbb{S}$'s odd half (the even grading $(1, X, P, W)$ forces the odd half into a mirrored weight pattern, admissible only at a distinguished contraction point $b_0 = 0$, where the odd diagonal squares vanish and only cross-products survive — a Grassmann-type induced structure), two candidate wall masses are tested.

The naive candidate $K = \Omega J_0$ ($J_0 = -L_{e_4}$, the gauge-compatible complex structure identified with hypercharge blockwise) is a **removable pure twist**: no gap, no wall modes. A composite mass built instead from K' — drawn from the four-dimensional antisymmetric, Ω-odd, gauge-covariant cell — localizes wall modes once a threshold parameter $\nu > 1$ is crossed, with eight modes forming two lepton and six quark real degrees of freedom.

The gate theorem: no operator is simultaneously Ω-odd and T_3-commuting, because T_3's eigenspaces *are* the chirality columns ($T_3 = \lambda J$ from §3.4). Any mass binding the two chirality columns necessarily breaks weak isospin — the Standard Model's own structural reason that fermion mass requires electroweak breaking, re-derived from the framework's own identity.

Forced chirality, by Schur's lemma. Color commutes with every operator defining the construction. The zero-mode projection is therefore color-equivariant, and cannot partially dismantle an irreducible complex triplet: the wall's quark sector is a whole **3** or $\bar{\mathbf{3}}$, and $\mathbf{3} \ncong \bar{\mathbf{3}}$. **The wall content $\{\mathbf{1}_{y_\ell}\} \oplus \{\mathbf{3}_{y_q}\}$ is necessarily chiral under the unbroken gauge group, independent of every spectral detail beyond the localization threshold's own existence.**

4.3 The orientation residue

All four orientation structures commute with the entire dynamics constructed above, and the dynamics provably cannot prefer among them: **the zero-energy wall sector is orientation-degenerate.** Whether the wall's chiral class is the Standard-Model-shaped one or its mirror is the one binary datum this stage's dynamics does not fix — not a late defect, but the polarization doctrine's own endpoint. This is the question Paper III inherits.

---

5. The protection theorem

5.1 The closed algebra and two dead theorem-candidates

Paper III audits the complete legal operator algebra available at this stage — the graded legality menu, the derived operators A_{sq} (squeeze) and A_{dil} (dilation), the wall couplings K, K', the structures Ω, J_0, and their block projectors — proving this list closed under multiplication (products cannot leave the partition).

Two plausible theorem-candidates are tested and die by explicit construction, reported as the record they are, not smoothed over: "realness protects the orientation" fails because a legal, central perturbation *is* itself an orientation, its sign a free knob; and A_{dil}'s candidacy as "the breaker" fails because its lepton-asymmetric part in fact anticommutes with J_0 globally — the correction that produces the trace lemma below.

5.2 Four mechanisms, jointly exhaustive

(i) The mode-reality lemma. Every symmetric-internal legal perturbation compresses to exactly zero on the wall — legality forces symmetric internals onto skew line-partners, and a skew operator's diagonal against a real bound profile vanishes identically. This alone removes A_{sq}, A_{dil}, Ω, and the entire symmetric commutant from the selection question.

(ii) The trace lemma. Every J_0-odd perturbation's alignment is exactly zero, at every threshold value, through every legal channel: the compression stays J_0-odd, the polar part of an odd operator is odd, and the trace of an odd operator vanishes by conjugation.

(iii) The per-line theorem. The framework's derived-sign odd operators are diagonal in one basis; their even products are therefore per-line scalars, structureless for alignment. The one candidate loophole — an odd product combining two derived signs — closes on direct inspection.

(iv) Sign-provenance and gauge-orbit closure. Every remaining alignable operator's sign is shown to be a **gauge** or **convention** choice, not a derived fact: the weak Weyl element flips Ω and K's signs exactly while leaving J_0 invariant; the T_3-conjugation orbit sweeps K''s entire cell at a fixed frequency, reaching $-K'$ at an exact, computed point. **J_0's own sign is the question, and is derived by nothing.**

5.3 The arrow-lock and the verdict

The framework's one genuinely derived sign — the thermal arrow, carried by A_{sq} — sits in the J_0-odd sector, hence is trace-locked away from the orientation by mechanism (ii), at every order: the even power A_{sq}^2 is a signless projector, and no combination both carries the arrow's sign and escapes J_0-oddness. **Time's direction provably cannot select the mirror.** This closes the negative half of what Paper III's charter calls the two-arrows conjecture — decisively, not provisionally: order-robustness is argued explicitly (a J_0-odd perturbation's second-order effect rides the coupling squared, hence signless; mixed terms carry a knob factor, hence remain a knob), so no perturbative order can smuggle a derived sign past this partition.

Verdict. Every channel available to derived structure is closed by its own theorem; the closures jointly exhaust the entire legal operator menu. The orientation of chirality is a **state-historical datum** — the framework's first genuinely initial-condition quantity — with the sole surviving author-candidate the settling stratum: non-quasi-free by its own definition, hence structurally outside everything this theorem covers.

---

6. The seam and the orientation-bearing sector

Paper III's verdict leaves one open door: the settling stratum, non-quasi-free and hence outside the protection theorem's jurisdiction by construction. Becoming's own text names the only algebraic home available for such a mechanism: the **zero-divisor structure** of $\mathbb{S}$ — the one structure necessarily mixing both Cayley–Dickson halves. Paper IV maps this structure combinatorially, asking exactly where, if anywhere, orientation information could live within it.

6.1 The tunnel lemma

For any two-term zero divisor A (an element $e_i \pm e_{8+u}$ admitting a nonzero annihilating partner), both diagonal blocks of $T_A = L_A^\top L_A$ are exactly flat — $N(A)\cdot$ Id on each half — so the entire three-band spectrum $\{0^4, N(A)^8, 2N(A)^4\}$ resides in the **half-mixing cross-block** alone: a zero divisor's distinguishing content is quantitatively a tunnel between the two halves. The kernel of L_A is exactly τ-balanced (every annihilated direction half-even, half-odd) yet commutes with neither τ nor the mediator's own complex structure — a genuinely entangled subspace, orthogonal to every grading the framework otherwise uses. Confirmed identically on the classical generator pair, three further discrete generators, and one genuine general-position anchor rotated off the basis skeleton entirely.

6.2 The algebra's PT, and the box-kites

$\varphi = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$, applied identically to both Cayley–Dickson halves, is an exact automorphism of $\mathbb{S}$ commuting with τ, conjugating L_{e_4} to $-L_{e_4}$ — read, on these verified facts, as the algebra's own **PT** operation, since every gauge-side orientation descends from the mediator's own sign.

The canonical two-term zero divisors number 84, on 42 index-planes ("assessors"); the raw annihilation graph — recovered from annihilation relations alone, with no external classification imported — decomposes into **seven components of twelve, uniform degree 4**: the box-kites. φ's character is uniform within each kite and splits the seven **3 PT-even + 4 PT-odd**, with no mixed kite. Each kite omits exactly one Fano (imaginary) direction — its **strut** — the same index absent from both halves; the correlation between a kite's PT-character and its strut's own φ-parity is exact, seven for seven: the three P-strutted kites (e_1, e_2, e_3) are PT-even; the four strutted on the mediator and the X-triple ($e_4; e_5, e_6, e_7$) are PT-odd.

6.3 What this establishes

The orientation-bearing sector of the zero-divisor structure is exactly the four PT-odd box-kites — 48 diagonals on four components, where choosing a diagonal is choosing a PT-orientation — while the three P-strutted kites are provably mirror-blind. This axis (the P/X distinction) is the same axis constituting the derived dynamics' entire internal content from §4–5 (A_{sq}'s menu-fraction-zero content) — the framework's flow-flavor and its seam-combinatorics share one axis, unarranged by anything in the construction.

This sharpens the settling conjecture to its most concrete form yet: settling selects the mirror if and only if its mechanism anchors in the PT-odd sector and breaks a diagonal symmetry there. Nothing in Paper IV asserts the positive half of this; it maps the terrain the surviving conjecture must cross.

---

7. Transport, quantization, and the carrier verdict

Paper V transports the seam note's combinatorial landscape to the gauge side, builds the owed fermionic quantization on it, discharges a standing mathematical condition the whole approach depends on, and tests Becoming's candidate mechanism to its conclusion.

7.1 Stage A: the bridge transport

Every assessor (i, u) satisfies $e_i \cdot e_u = \pm e_S$ for its kite's strut S, exactly, across all 42 planes. The mediator kite's six assessors are exactly the three J_0-partner pairs $\{1, 5\}, \{2, 6\}, \{3, 7\}$ — entirely within-line — while every other kite is entirely cross-line. Among the seven generator left-multiplications L_{e_i}, only L_{e_4} commutes with the full color group. The real unit never appears in any two-term zero divisor, and combined with e_4's within-kite-only appearances, the **lepton line is structurally undecoratable**: no two-term diagonal can carry a sign on it.

A composed identity — the algebra's PT sending $e_4 \to -e_4$, transported through the directed strut fibration, matched against the wall-side global flip's own action — closes exactly (quark-block-scoped for convention-independence; the lepton-inclusive global form holds too, once one embedding convention is pinned, the apparent failure traced to a catalogued convention divergence between two internally-consistent implementations, not a mathematical defect).

Verdict [anomaly in the frozen taxonomy]: transport is canonical and theorem-forced on **exactly one channel** — the mediator fiber's single bit, mapping onto $\pm J_Q$ — with every other channel doubly excluded (gauge-incompatible; remnant-swept wherever anything remains to sweep). The lepton-relative sign sits outside every invariant space tested. One bit carried; everything else provably silent.

7.2 Stage B: the CAR construction

Eight self-adjoint Clifford generators, built from octonion left-multiplications on the wall's eight real modes, satisfy the CAR relation exactly. The mediator-partner pairing gives four independent fermionic number operators with joint spectrum exactly binomial, $\{-4, -2^4, 0^6, +2^4, +4\}$ — confirming the sixteen-dimensional Fock space directly from the operator spectrum. The four orientation classes sit at the four maximally-polarized Fock states, $(\ell, q) = (\pm 1, \pm 3)$, each the exact one-dimensional joint kernel of its own orientation's four lowering operators and of no other's. Since the wall Hamiltonian vanishes identically, $\rho_{\text{wall}} = \text{Id}/16$ at every temperature: the thermal path is orientation-blind at every β, its zero-temperature limit the mixed average rather than any vacuum.

7.3 Stage C: the standing condition

Depth-change is proven, unconditionally, to require non-Bogoliubov dynamics: symplectic eigenvalues are invariant under every symplectic transformation, and the framework's depth coordinate is built from exactly this invariant. Separately, each quasi-free family member is confirmed a fixed point of the framework's own derived flow — a narrower, first-hand fact establishing that whichever branch (dissipative, restriction-based, or nonlinear-unitary) Stage D's mechanism class occupies, it sits outside symplectic implementation and hence outside Paper III's audited jurisdiction entirely. This is the jurisdictional guarantee Stage D needs; it is not a claim that the registered instruments extend automatically to whatever mechanism is found there, which is checked separately as each ingredient is extracted.

7.4 Stage D: the anchor question

Becoming's own text names the candidate mechanism class verbatim: *a singular scaling limit anchored at the zero-divisor locus, degenerating along exactly the directions where multiplication already fails to invert.* Extracted into three explicit operators for an anchor A — L_A, $T_A = L_A^\top L_A$, and P_{kernel}, the projection onto T_A's four-dimensional tunnel — the construction is tested for whether it genuinely imports directional information from a specific point of the locus, or only ever reflects the locus's own overall symmetry.

Anchored at one specific zero divisor, all three ingredients show substantial, robust commutators with the framework's full symmetry algebra: genuine directional content. **Averaged over the entire locus** (computed exactly, via the diagonal symmetry-commutant, not by sampling): $\langle T_A \rangle = 2 \cdot \mathrm{Id}$ and $\langle L_A \rangle = 0$ exactly — direction-free, contentless for orientation. The averaged kernel projector is not scalar, however: its exact spectrum $\{0, 0, (2/7)^{14}\}$ retains the algebra's own scalar/vector type grading even while direction-blind — refining, not confirming, an earlier "total vacuum" expectation.

The scalar-avoiding kernel theorem, proven by a positivity argument and verified non-statistically on multiple discrete anchors and one genuine general-position anchor: P_{kernel} annihilates the real-unit sector for *every* zero divisor on the locus, not merely on average.

The coupling resolution. The anchored mechanism's coupling to the gauge side is not linear (two direct alignment tests give exact zero, both for structural reasons, verified against a working positive control) but quadratic, residing in the tunnel itself: flipping an anchor's sign leaves L_A's diagonal blocks identical while flipping the cross-block — the tunnel — to its exact negative. The full chain — anchor $\to$ tunnel direction (quadratic) $\to$ directed strut (exact) $\to$ wall orientation class (§7.1's consistency square) — is independently verified at every link.

Verdict. The mechanism class has exactly two kinds of member: **direction-free**, orientation-contentless but not structureless (the type-split residue survives); and **anchored**, contentful, carrying the mirror bit through the fully verified chain above to $\pm J_Q$. There is no third road within this class. **Nothing in Becoming's text, the redistribution paper, or anywhere else in the corpus supplies which point of the locus anchors.** Settling, as this mechanism class specifies it, is a **carrier** of orientation, not an **author** of it. Whether the framework structurally *cannot* supply an anchor-selector is a stronger claim this stage does not make — Stage C's own discharge places the settling stratum outside Phase 4's audited jurisdiction, and an anchor-selector would live in that same un-audited territory; nothing proven forbids that territory from containing one. Any candidate anchor-supplier would itself have to live outside everything audited here — which is exactly where a future document would have to look.

---

8. Synthesis: forced, protected, carried

The complete arc, stated once as a single narrative:

A single mediator direction and the idempotent it fixes force the Standard Model's gauge group with exact charges. The same algebra's own dynamics, applied to that matter content, forces a chiral wall sector — necessarily, by Schur's lemma acting on an irreducible triplet — while leaving exactly four gauge-compatible orientation classes among which the dynamics is provably indifferent. A closed, exhaustive theorem then proves that nothing derivable from the framework — no legal operator, no perturbative order, not even the framework's own thermal arrow — can select among those four classes: the orientation is a genuine initial-condition datum, not a gap awaiting a cleverer construction. The only structure capable of hosting a non-derived selector is the zero-divisor locus, mapped combinatorially into an orientation-bearing sector of exactly four box-kites out of seven. Testing the one mechanism class the framework's own text proposes for this locus: it can carry the orientation bit, faithfully, through a fully verified algebraic chain — but the framework supplies no rule for which point of the locus is occupied.

Chirality is forced. Its mirror is protected from every derived mechanism. The one thing that could carry that mirror is real, mapped, and tested — and it is handed no address.

Two readings of this verdict are both live, and nothing in the corpus adjudicates between them. Either this is among the deepest results the program has produced — a genuine proof that a physical orientation can be, in the strict sense, an unprovable initial condition — or it is a precise, honest way of saying "not derivable from the tools assembled here," with the true mechanism simply not yet found.

9. What remains open

The anchor-supplier question (registry item REG-13). Any selector for the zero-divisor locus's specific anchoring point would have to live entirely outside Paper III's audited, symplectic jurisdiction — a located address, not a proof of non-existence, and not yet a candidate. Since these papers closed, the settling program (Program ω; the gate note and interpretation session) has opened and produced its first results: the anchored mechanism class is now derived to split into three dynamical characters by an exact composition rule, with the 24 extreme-class anchors carrying a conserved four-dimensional core — sharpening REG-13 into a typed choice and supplying the first anchor-property with a dynamical signature, without supplying the anchor.

The preservation lemma (REG-14). Needed to upgrade Foundation's non-Bogoliubov trichotomy (settling is dissipative, open-system, or restriction-based) from a citation to a theorem: that unitaries mapping the quasi-free family into itself are necessarily of Bogoliubov type. Supplied nowhere in the corpus.

The lepton-relative sign. Carriable, if at all, only at the price of breaking the $\mathfrak{so}(3)$ remnant of color — the deepest symmetry surviving every prior stage of this program. Whether nature pays that price is entirely untouched by anything above.

Reading B, standing. Every claim in this paper concerns internal chirality on the algebra's own derived scale line. Its identification with spacetime's γ^5 is conditional on a spacetime program not yet begun; Reading A (the weak sector merely *contains* a chirality axis, with no further spacetime claim needed) remains live and undecided against it.

The e_R/generation boundary. No half of the matter content constructed here is a complete generation (§3.5); this is unmoved by anything in §4 onward, and is not a target of the chirality program as such.

None of these are computations waiting to run — each is gated behind programs either not yet started or only now beginning. A synthesis that states them as open is complete; one that waited for them to close would never be written.

10. Verification and provenance

Every claim in this paper was independently verified in two implementations at the level of Papers I–V — separate algebraic constructions, separate derivation-system assemblies, separate stabilizer and commutant extractions — with results diffed and reconciled, not merely compared for plausibility. Papers I, IV, and V each closed after at least one round of hostile review, with review targets and their resolutions recorded in each paper's own closing note; Paper III closed after two rounds. Two theorem-candidates (§5.1) and one non-zero-divisor anchor used without checking (§7.4) are reported in the source papers as errors caught and corrected, not smoothed into the record — the error ledgers accompanying Papers II and V in particular list every catch, on both sides of the collaboration, as a matter of methodological record. This paper reproduces none of that verification detail; it exists to state the finished result cleanly, with Papers I–V as the place every claim above can be checked against its own proof.

References

- Paper I: *The Standard Model's Structure from* $(\mathbb{S}, \tau)$*: Gauge Group, Exact Charges, a Native Chirality Mechanism, and the Proven Boundary of* $J_3(\mathbb{O})$ (Revision 1.3, closed).
- Paper II: *Forced Chirality: The Derived Fermionic Dynamics of* $(\mathbb{S}, \tau)$*, the Domain-Wall Sector, and the Orientation Residue* (the Phase 3 report).
- Paper III: *The Protected Mirror: Why Nothing Derived Can Choose the Orientation* (the Phase 4 report, Revision 1.1, closed).
- Paper IV: *The Seam and the Mirror: Zero Divisors, the PT Automorphism, and the Orientation-Bearing Sector* (Revision 1.1, closed).
- Paper V: *The Bridge, the States, and the Dial: Phase 5, Stages A–D* (Revision 1.0, closed).
- Companion documents: *Becoming* (the zero-divisor conjecture's textual source); *The Redistribution Operator* (the tunnel's associator mechanism); Foundation and Keystone (for $(\mathbb{S}, \tau)$ and the quantization chain).
- Dubois-Violette, M., "Exceptional quantum geometry and particle physics," *Nucl. Phys. B* 912 (2016) 426–449.
- Todorov, I. & Dubois-Violette, M., "Deducing the symmetry of the standard model from the automorphism and structure groups of the exceptional Jordan algebra," *Int. J. Mod. Phys. A* 33 (2018) 1850118, arXiv:1806.09450.
- Boyle, L., "The standard model, the exceptional Jordan algebra, and triality," *J. Math. Phys.* 67 (2026) 071701.
- Baez, J. & Schwahn, P., "The Standard Model Gauge Group from the Exceptional Jordan Algebra," arXiv:2606.15235.
- Krasnov, K., "$SO(9)$ characterisation of the standard model gauge group," arXiv:1912.11282.
- Moreno, G., "The zero divisors of the Cayley–Dickson algebras over the real numbers," *Bol. Soc. Mat. Mexicana* (3) 4 (1998), 13–28.
- de Marrais, R. P. C., "The 42 Assessors and the Box-Kites They Fly," arXiv:math.GM/0011260 (2000).

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.